

# Stagehand Phase Evidence

Current checkpoint: public Site v99, verified 2026-09-03.

## Project

Stagehand is a WebMCP-native 2D animation studio for editable paper-cutout scenes. The current cut supports rigged characters, imported image props, per-asset visual direction, camera work, captions, non-voice audio cues, storyboard beats, multi-scene Preview, and deterministic WebM export.

Private source: [github.com/swaymun/stagehand-animation-studio](https://github.com/swaymun/stagehand-animation-studio)

Source HEAD: 216672e1b4dfd715bbed2991815f4f84860148ba

Deployed runtime: 216672e1b4dfd715bbed2991815f4f84860148ba

Public Site: [stagehand-animation-studio.saimun-h-shahee.chatgpt.site](https://stagehand-animation-studio.saimun-h-shahee.chatgpt.site)

Sites version: 99

## Acceptance loop

Implement - local tests - Sites deployment - hosted test - UI roast - fixes - second review.

## Implemented at this checkpoint

- Revision-safe WebMCP mutations with optional expectedRevision and structured REVISION\_CONFLICT results.
- Idempotent WebMCP mutations with optional idempotencyKey replay protection.
- Structured per-asset role, treatment, silhouette, palette, and direction notes with human and agent parity.
- Asset treatment choices reach canvas, thumbnails, Preview, and WebM; new assets also receive valid style defaults immediately.
- Motion actions show affected character and bounded duration; pose presets remain a separate group.
- Human audio cue Start/End editors share bounded timing updates with the agent cue editor.
- Animate and Storyboard are the only top-level editor modes; Preview is a compact review action and overflow actions keep infrastructure chrome out of the workspace.
- The timeline defaults to semantic camera, pose, prop, dialogue, music, and SFX event bands; Show details restores raw keyframe controls.
- Responsive project and Inspector drawers preserve a readable shell at phone widths.
- At split-browser widths up to 1200 px, the Inspector is closed by default and opens as an overlay drawer while the stage and project rail retain working width.
- The editor preserves a 16:9 canvas frame, removes permanent canvas debug labels, and wraps compact timeline actions below 900 px so scene titles and controls remain readable.

- The compact Inspector has a visible close control inside its own drawer header; the responsive smoke gate covers the full open/close lifecycle.
- A Sol-powered Codex run used the public page's live WebMCP tools to apply and validate an Alice/Bob coupon gag without a separate MCP connection or TOML change.

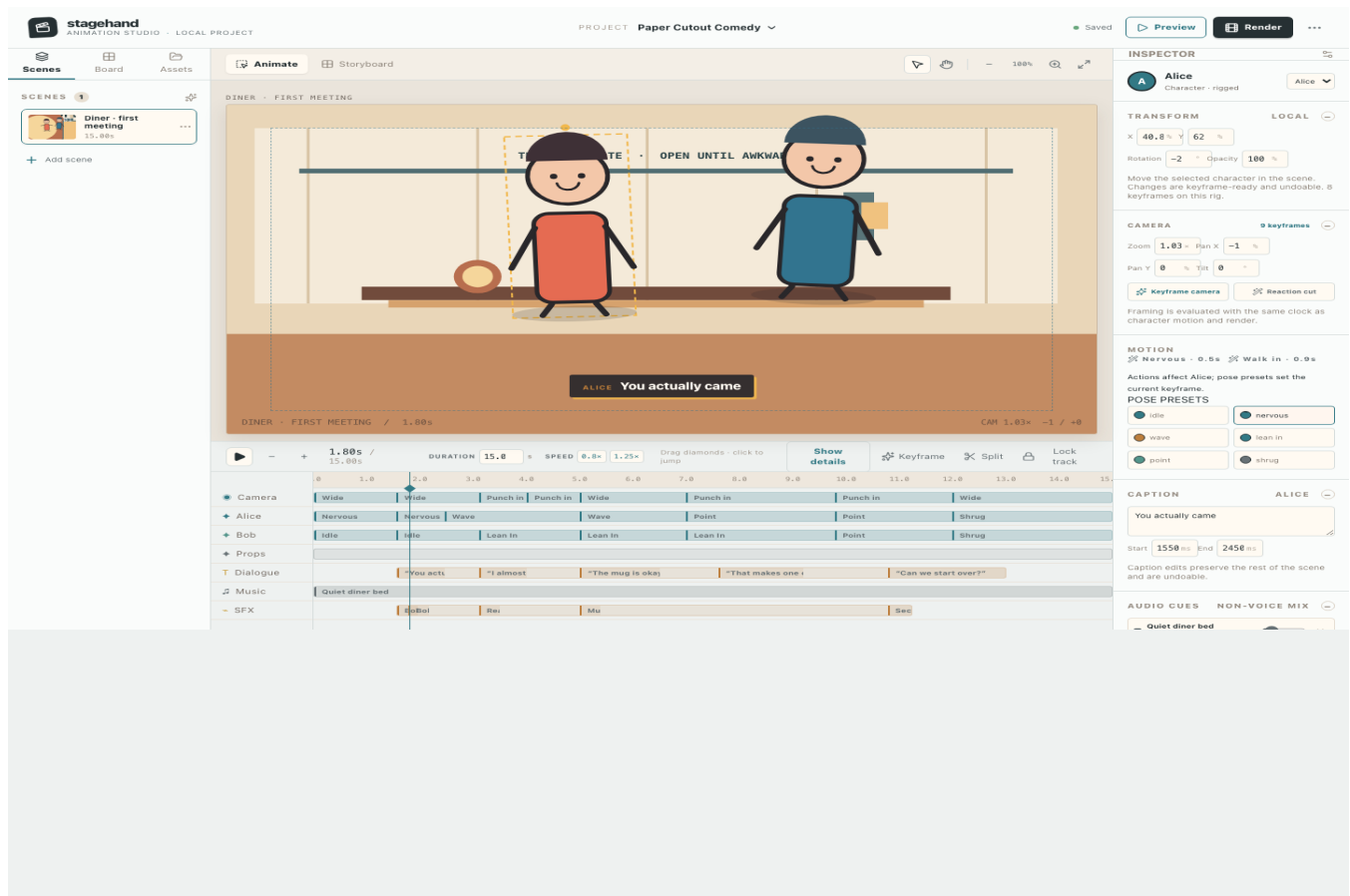
## Verification

Local gate:

```
npm run format
npm run lint
npm run build
git diff --check
npm run smoke
npm run smoke:native
```

Smoke passed: 52 tools, 39 guarded mutations, a 15-second six-beat starter, synchronized timing, asset treatments, import/recovery including empty collections, Preview, valid PNG/WebM signatures, native WebMCP registration, and zero errors.

# Animate editing workspace



Hosted Playwright capture at 1440 x 960. The focused editor keeps artwork dominant, with semantic event bands and raw-keyframe details available on demand.

# Assets style editor

The screenshot shows the Stagehand Animation Studio interface. The main window displays a scene titled "DINER - FIRST MEETING" with two characters, Alice and Bob, in a diner setting. Alice is a red character and Bob is a blue character. The interface includes a left sidebar with asset lists, a top navigation bar, and a right sidebar with the Asset Inspector. The Asset Inspector shows details for "Alice - rigged" and "Bob - rigged" assets, including their placement and style settings. The bottom of the interface features a timeline and a storyboard view.

**Assets**

- Alice - rigged
  - Rigged character - starter - ...
  - HERO - PAPER
  - CORAL - DINER TEAL
- Bob - rigged
  - Rigged character - starter - ...
  - HERO - PAPER
  - DINER TEAL - WARM PAPER
- Diner background
  - Background - starter - ON STAGE
  - ENVIRONMENT - PAPER
  - WARM PAPER - DINER TEAL
- Coffee mug
  - Prop - starter - LIBRARY
  - ACCENT - FLAT-COLOR
  - MUSTARD - AMBER

**ADD PLACEHOLDER**

- + Rigged character
- + Background
- + Prop
- + Audio
- Import character art
- Import pose sheet - auto
- Import background
- Import prop

Import character art to bind it to the selected rig; poses and keyframes stay editable. Pose-sheet imports auto-detect a wide four-column image and fall back to single art when needed.

**ASSET INSPECTOR**

**Alice - rigged**  
RIGGED CHARACTER - STARTER

**PLACEMENT** UNBOUND  
Available for rig binding.

**STYLE** PAPER  
Hero Clear Silhouette

[Edit style](#)

Asset identity and art direction live here. Motion and rig controls return when you select Scenes.

**Storyboard**

DINER - FIRST MEETING / 1.00s

1.00s / 15.00s DURATION 15.0 s SPEED 0.8x 1.25x

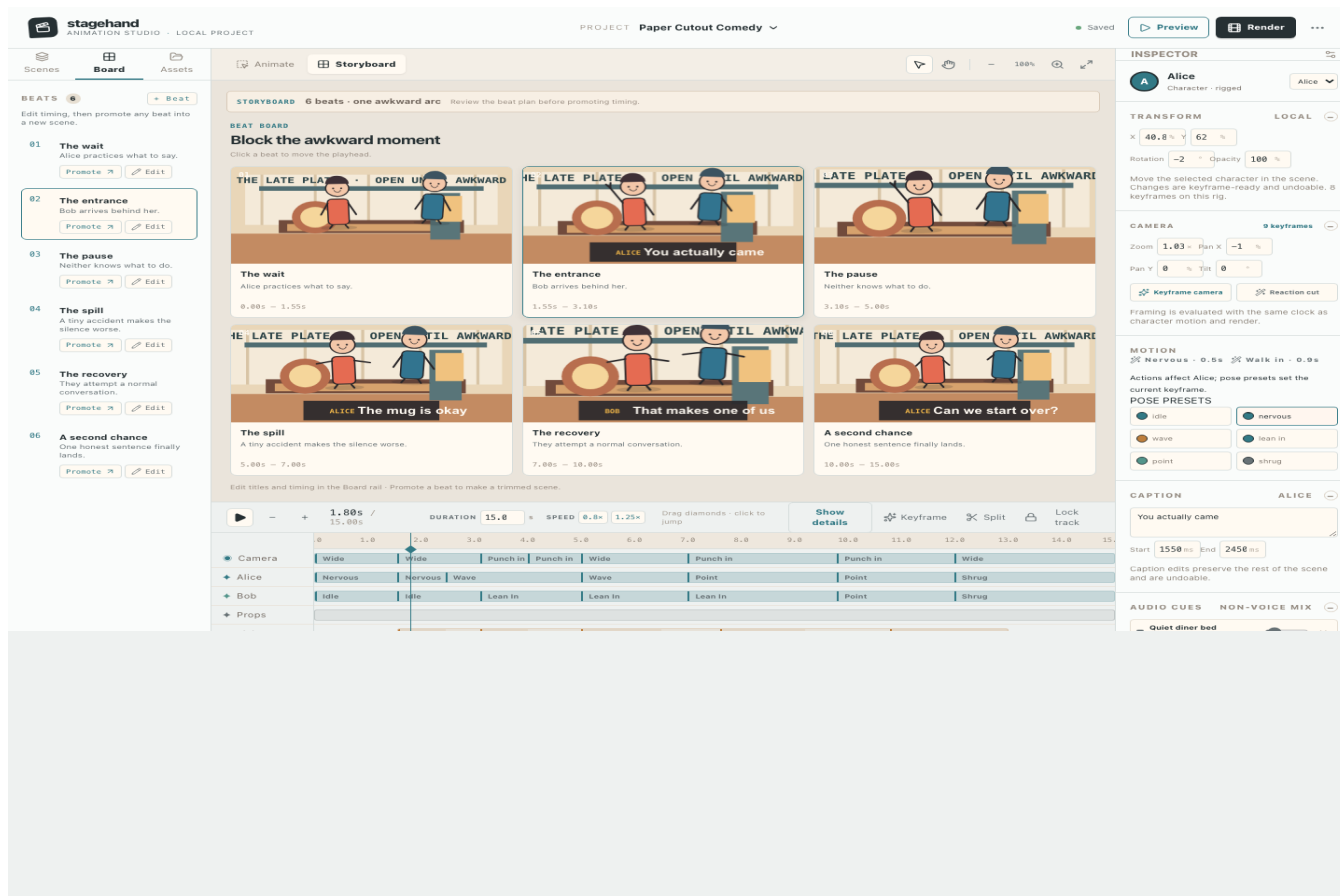
Drag diamonds - click to jump

Show details Keyframe Split Lock track

|          | 0               | 1.0             | 2.0             | 3.0             | 4.0              | 5.0             | 6.0             | 7.0             | 8.0             | 9.0             | 10.0            | 11.0            | 12.0            | 13.0            | 14.0            | 15.0            |
|----------|-----------------|-----------------|-----------------|-----------------|------------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|-----------------|
| Camera   | Wide            | Wide            | Punch in        | Punch in        | Wide             | Punch in        | Punch in        | Punch in        | Punch in        | Punch in        | Punch in        | Punch in        | Punch in        | Punch in        | Punch in        | Wide            |
| Alice    | Nervous         | Nervous         | Wave            | Wave            | Wave             | Wave            | Wave            | Wave            | Wave            | Wave            | Wave            | Wave            | Wave            | Wave            | Wave            | Wave            |
| Bob      | Idle            | Idle            | Lean In         | Lean In         | Lean In          | Lean In         | Lean In         | Lean In         | Lean In         | Lean In         | Lean In         | Lean In         | Lean In         | Lean In         | Lean In         | Lean In         |
| Props    |                 |                 |                 |                 |                  |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |                 |
| Dialogue |                 |                 | "You acti       | "I almost       | "The mug is okay | "That makes one | "That makes one | "That makes one | "That makes one | "That makes one | "That makes one | "That makes one | "That makes one | "That makes one | "That makes one | "That makes one |
| Music    | Quiet diner bed | Quiet diner bed | Quiet diner bed | Quiet diner bed | Quiet diner bed  | Quiet diner bed | Quiet diner bed | Quiet diner bed | Quiet diner bed | Quiet diner bed | Quiet diner bed | Quiet diner bed | Quiet diner bed | Quiet diner bed | Quiet diner bed | Quiet diner bed |
| SFX      |                 |                 | Bobol           | Rei             | Mu               | Mu              | Mu              | Mu              | Mu              | Mu              | Mu              | Mu              | Mu              | Mu              | Mu              | Mu              |

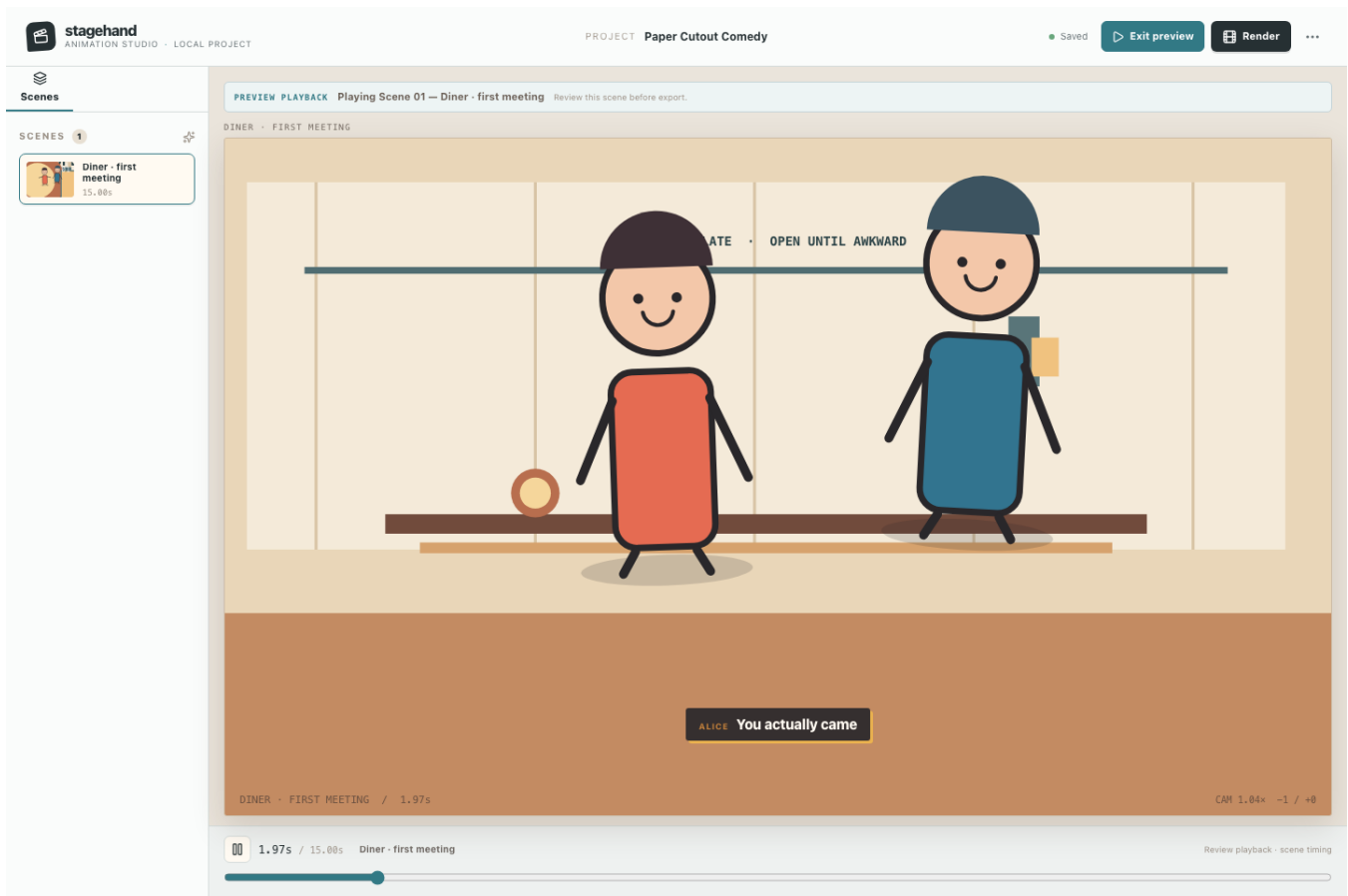
Hosted Playwright capture at 1440 x 960. Scannable asset rows stay in the rail while the selected asset's style and placement context live in Inspector.

# Storyboard mode with Board rail



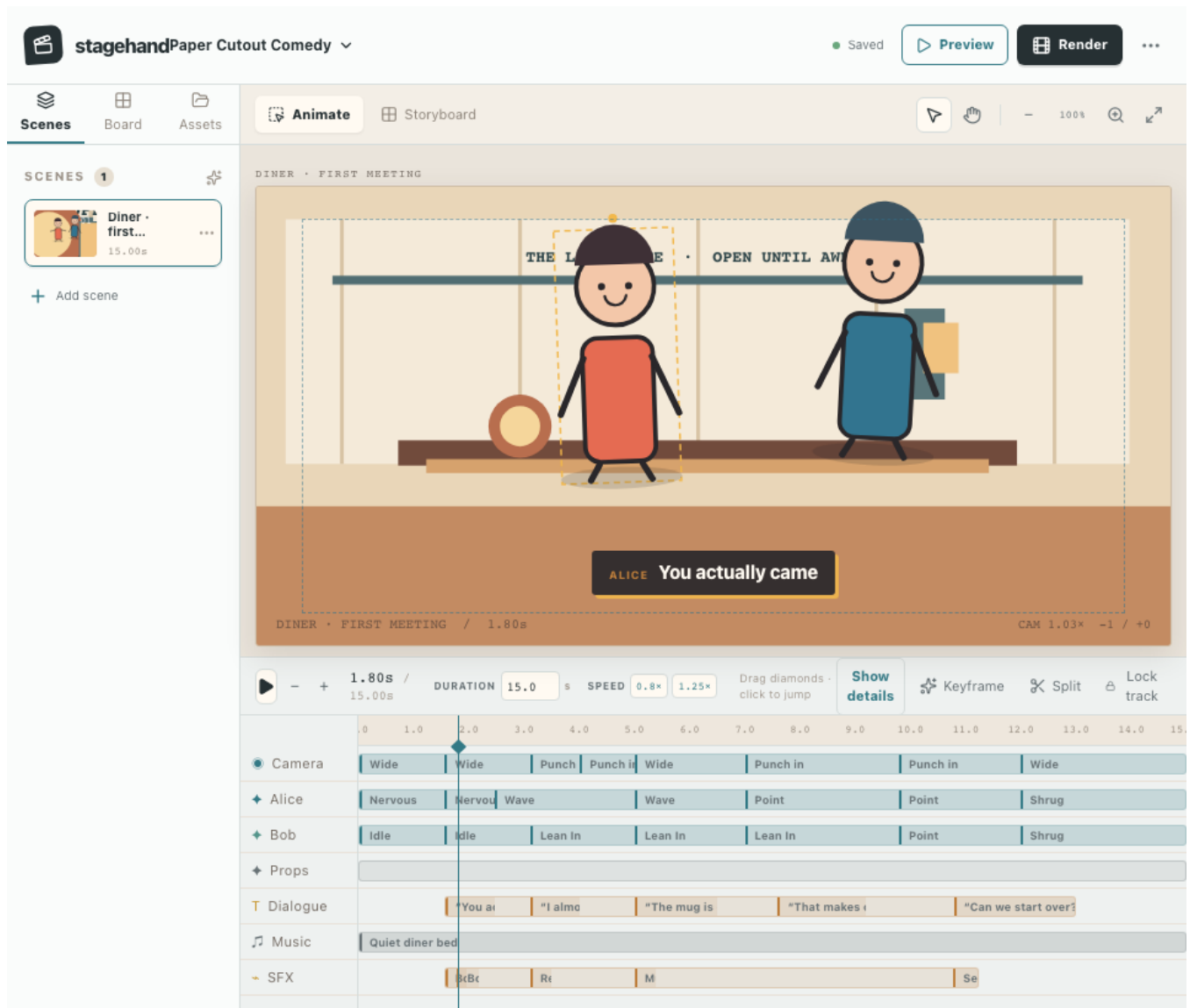
Hosted Playwright capture at 1440 x 960. Six beat cards make the longer arc inspectable while the selected mode and project rail stay synchronized.

# Review-first Preview player



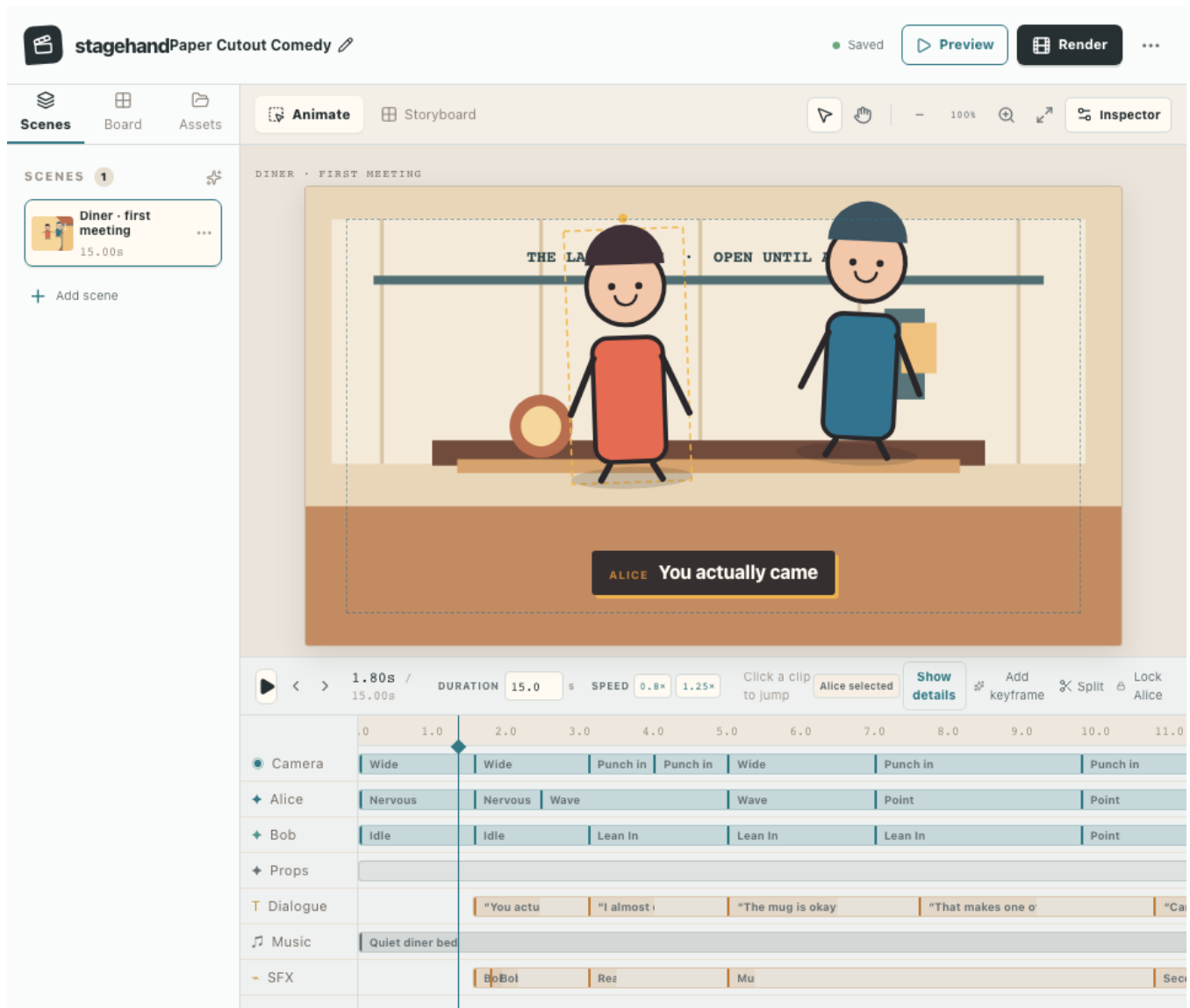
Hosted Playwright capture at 1440 x 960. Scene-only context, transport, and a compact scrubber remain; the editable timeline grid and Inspector are hidden.

# Split-browser compact editor



Hosted Playwright capture at 960 x 820. The Inspector becomes a closed-by-default drawer, leaving the scene rail and artwork usable beside a ChatGPT or Codex conversation.

# Final split-browser compact editor



Hosted Playwright capture at 960 x 820. The corrected 16:9 stage, readable scene title, visible Inspector action, and compact timeline controls are all covered by the hosted responsive gate.



## Roast findings and fixes

- Imported props lacked direct human transform editing. Added X/Y/Scale/Rot controls sharing the keyframe command path.
- Four prop transform fields initially collided with the asset-list flex rule and truncated. Replaced it with an explicit two-column grid; computed field boxes are 57 px wide in two rows.
- Preview carried too much editing chrome. Converted it into a review-first player with scene-only context and Exit preview.
- Timeline diamonds had 7 x 7 px hit targets. Enlarged the buttons to 22 x 22 pixels while preserving the visual diamond.
- Agent retries could duplicate successful writes. Added optional idempotency-key replay protection and a hosted smoke assertion.
- Switching top-level Storyboard mode could leave Assets selected. Synced mode selection to its companion project rail.
- Character transform inputs lacked explicit accessible names. Added dynamic X/Y/rotation/opacity labels and a smoke check.
- Asset briefs carried style intent only as prose. Added structured role, treatment, silhouette, palette, and direction fields; later made those treatments visible in the canvas and manifest placement cues.
- Motion action buttons were visually mixed with pose presets. Added explicit durations, affected-character copy, and a Pose presets label.
- Agent frame-rate-only render updates reset an existing 1080p choice. Preserved the current resolution when the preset is omitted and added smoke coverage.
- Preview still exposed global project-editing controls. Hid Settings, Import, Export JSON, and rename while retaining WebM and PNG review actions.
- Scene titles competed with the ready badge and were visually shortened. Used a two-line title clamp and positioned the badge outside the text flow.
- The smoke gate did not exercise recovery, modal semantics, or cross-scene playback. Added Help open/close, reload recovery, and sequence Preview transition checks to local and hosted runs.
- The Inspector was a long scroll dump, human scrubbing could leave agent reads one event behind, and keyboard stepping was unreliable after scrub focus. Added native collapsible groups, shared project-view synchronization, a focused stage target, and smoke coverage.
- Export checks only proved that downloads were non-empty. Added binary signature assertions for WebM/EBML and PNG outputs in local and hosted smoke runs.
- Experimental Chromium exposed navigator.modelContext while the app only checked document.modelContext. Added a compatibility fallback and a native registration smoke gate covering all 52 tools plus valid and invalid calls.
- Preview still read like the editor because raw timeline, guides, and status toast survived into review. Added a compact scene/timing scrubber and hid editing-only surfaces.

- The asset rail carried too much editing UI. Kept it scannable and moved the full style/brief editor into the wider Inspector.
- Header/sidebar implementation details competed with the artwork. Made Preview and Render primary actions, moved secondary actions under More, and removed permanent validation/tool-count/starter cards.
- The timeline's first read was keyframe noise. Added semantic event bands with an explicit Show details disclosure for precise edits.
- The desktop shell could force phone-width overflow. Removed the global minimum width and added responsive project/Inspector drawers with a readable-width timeline.
- In a split ChatGPT/Codex browser pane, the permanent Inspector consumed the stage's working width. Moved it to a closed-by-default overlay drawer at  $\leq 1200$  px, hid the redundant project-drawer action while the rail remains visible, and added a checked-in 960 px regression gate.
- The split-pane stage could stretch toward 2:1 and disagree with the 16:9 export frame. Added a responsive centered 16:9 canvas frame, removed permanent debug metadata, and added a ratio assertion.
- At narrower in-app widths, timeline actions and scene titles competed for the same compact row. Wrapped timeline actions below 900 px, contained toolbar overflow, widened the internal timeline surface, and overlaid the scene kebab action.
- The compact Inspector's open trigger sits behind the drawer after opening. Added a visible close button inside both Inspector header variants and a hosted open/close lifecycle assertion.

## Limits

The regular hosted interaction suite remains a Playwright fallback, while a Sol-powered Codex run directly exercised the public page's WebMCP tools in the in-app browser and verified the Alice/Bob coupon gag. The Luna-bound main task cannot call browser WebMCP; supported Sol/Terra tasks can. No separate MCP server connection, TOML entry, or restart was needed. Native Chromium smoke complements that live run. TTS, voice cloning, dialogue recording, phoneme extraction, and lip-sync remain out of scope.